Name_____

Calculus: Limits, Continuity and the Intermediate Value Theorem

1. Calculate:

$\lim_{h\to 0} \dfrac{\tan(x+h)-\tan x}{h}$ at $x=\dfrac{\pi}{2}$

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2. Let $g(x)=\frac{x^2-9x+14}{2x^2-11x+14}$. Determine and describe all discontinuities. State an equation of horizontal asymptotes, if any.

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3. Is $f(x)=\begin{cases} -x+1, x\le 0 \\ x-1, x>0 \end{cases}$ continuous at $x=0$? Explain using limits.

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4. Find all values of *k* so that f(x)=$\begin{cases} k\sqrt{x+1}, & 0\le x\le 3 \\ 5-x, & 3<x\le 5 \end{cases}$ is continuous at $x=3$.

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5. We wish to find the values of a and b so that the function with equation

$$f(x)=\begin{cases} x^2+ax+b, & x\le 1 \\ x+1, & 1<x\le 3 \\ x^2+ax+b, & x>3 \end{cases}$$

is continuous for all x.

A <u>continuous</u> function $f(x)$ on $[a, b]$ takes on every value between $f(a)$ and $f(b)$. This means . . .
$f(x) = k$ must have at least one solution if k is between $f(a)$ and $f(b)$. (textbook pg. 83)

Apply the Intermediate Value Theorem to answer each of the following:

6. Show that the polynomial function $f(x) = x^3 + x + 1$ has a value of zero between -1 and 0.

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7. Show that the function $x^2 - 4x + 2$ intersects the x-axis on the interval [0, 2].
   Can the same be said for the function $\frac{2x-3}{x-1}$? Explain.

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8. A function g is continuous on the closed interval [-1, 1].
   Values for g are given in the table to the right.

x	-1	0	1
$g(x)$	0	c	1

   For what value of c will the equation $g(x) = -\frac{1}{2}$ have at least two solutions?

   (A) 1 (B) $-\frac{1}{2}$ (C) 0 (D) -1 (E) $\frac{1}{2}$

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9. The function f is continuous for all real numbers with $f(-3) = -5$ and $f(1) = 2$.
   If f has exactly one zero, then its x-value could be what value? Why?

   (A) 1 (B) -4 (C) 4 (D) -3 (E) -1